

The Ultimate Web Development Handbook



Typography in CSS

Typography is how text appears on a web page — its **font**, **size**, **spacing**, **alignment**, **weight**, and overall readability. Good typography improves user experience and design quality.

Basic Text Properties

1. `font-family`

Sets the typeface for your text.

```
body {  
  font-family: Arial, sans-serif;  
}
```

- You can specify a list of fallback fonts.
- Always end with a **generic family** like `sans-serif`, `serif`, or `monospace`.

Common font stacks:

```
font-family: "Helvetica Neue", Helvetica, Arial, sans-serif;  
font-family: Georgia, 'Times New Roman', serif;  
font-family: 'Courier New', Courier, monospace;
```

2. `font-size`

Controls the size of the text.

```
h1 {  
  font-size: 36px;  
}
```

You can use units like `px` , `em` , `rem` , `%` .

```
p {  
  font-size: 1.2rem;  
}
```

3. `font-weight`

Defines the boldness of text.

```
strong {  
  font-weight: bold;  
}
```

You can use keywords like `normal` , `bold` , or numeric values like `100` , `400` , `700` , `900` .

4. `font-style`

Sets text to normal, italic, or oblique.

```
em {  
  font-style: italic;  
}
```

5. **text-align**

Aligns text horizontally.

```
h2 {  
  text-align: center;  
}
```

Values: **left** , **right** , **center** , **justify**

6. **line-height**

Controls the space between lines of text.

```
p {  
  line-height: 1.6;  
}
```

This improves readability, especially for paragraphs.

7. **letter-spacing**

Controls space between characters.

```
h1 {  
  letter-spacing: 2px;  
}
```

8. **word-spacing**

Controls space between words.

```
p {  
  word-spacing: 5px;  
}
```

9. text-transform

Changes the case of text.

```
.upper {  
  text-transform: uppercase;  
}  
  
.lower {  
  text-transform: lowercase;  
}  
  
.capitalize {  
  text-transform: capitalize;  
}
```

10. text-decoration

Controls underlining, overlining, and line-through.

```
a {  
  text-decoration: none;  
}
```

Using Google Fonts

To use custom fonts, you can load them from [Google Fonts](#).

HTML

```
<link href="https://fonts.googleapis.com/css2?family=Roboto&display=swap" rel="stylesheet">
```

CSS

```
body {  
  font-family: 'Roboto', sans-serif;  
}
```

Summary

Typography affects the readability and tone of your website. Key things to remember:

- Use `rem` for font sizing to keep things scalable.
- Set appropriate `line-height` and `font-family` for comfortable reading.
- Align and style text to match your design's personality.

Backgrounds and Borders in CSS

CSS allows you to customize how elements **look and feel** by adding backgrounds and borders. You can apply colors, images, gradients, and control borders precisely around elements.

Background Properties

1. `background-color`

Sets a solid color behind an element.

```
div {  
  background-color: lightblue;  
}
```

2. `background-image`

Adds an image as the background.

```
body {  
  background-image: url('background.jpg');  
}
```

You can use local images or remote URLs.

3. background-repeat

Controls if the background image repeats.

```
background-repeat: repeat;      /* Default */
background-repeat: no-repeat;
background-repeat: repeat-x;    /* Only horizontally */
background-repeat: repeat-y;    /* Only vertically */
```

4. background-size

Sets the size of the background image.

```
background-size: cover;        /* Fill container and crop */
background-size: contain;      /* Fit image without cropping */
background-size: 100px 200px;  /* Custom dimensions */
```

5. background-position

Positions the background image within the element.

```
background-position: center;
background-position: top right;
background-position: 50% 50%;
```

6. background-attachment

Controls scroll behavior.

```
background-attachment: scroll;   /* Default */
background-attachment: fixed;   /* Stays in place during scroll */
```

7. Shorthand Property: **background**

You can combine all background properties in one line.

```
div {  
  background: url('bg.jpg') no-repeat center center / cover;  
}
```

Border Properties

1. **border-width**

Sets the thickness of the border.

```
div {  
  border-width: 3px;  
}
```

2. **border-style**

Defines the style of the border.

```
border-style: solid;      /* Common */  
border-style: dashed;  
border-style: dotted;  
border-style: double;  
border-style: none;
```

3. `border-color`

Sets the color of the border.

```
border-color: darkgray;
```

4. Shorthand: `border`

You can combine width, style, and color.

```
div {  
  border: 2px solid #333;  
}
```

5. Individual Sides

```
border-top: 1px solid black;  
border-right: 2px dashed red;  
border-bottom: none;  
border-left: 3px dotted green;
```

6. `border-radius`

Rounds the corners of an element.

```
button {  
  border-radius: 10px;  
}
```

You can also use percentages to make circular shapes:

```
img {  
  border-radius: 50%; /* Perfect circle for square images */  
}
```

Summary

- Use background properties to add color, images, and gradients.
- Borders help separate and define content.
- Use `border-radius` to soften corners and create modern designs.

Margin and Padding in CSS

Margin and **Padding** are two of the most commonly used properties in CSS to control **spacing** around elements. They are part of the CSS **Box Model** and play a crucial role in layout and visual structure.

Padding vs Margin

Property	Affects	Where the space appears
<code>padding</code>	Inside the element	Between the content and the border
<code>margin</code>	Outside the element	Between the element and others

Padding

Apply space inside the border, around the content.

```
.box {  
  padding: 20px;  
}
```

This adds 20px space inside all four sides of the `.box`.

Individual sides

```
padding-top: 10px;  
padding-right: 15px;
```

```
padding-bottom: 10px;  
padding-left: 15px;
```

Shorthand

```
padding: 10px 15px 10px 15px; /* top right bottom left */  
padding: 10px 15px;          /* top-bottom | right-left */  
padding: 10px;               /* all sides */
```

Margin

Adds space outside the border of an element.

```
.card {  
  margin: 30px;  
}
```

This separates the `.card` from nearby elements.

Individual sides

```
margin-top: 20px;  
margin-right: 0;  
margin-bottom: 20px;  
margin-left: auto;
```

Shorthand

```
margin: 20px 40px 20px 40px; /* top right bottom left */
margin: 20px 40px;           /* top-bottom | right-left */
margin: 0 auto;              /* top-bottom: 0, left-right: auto (used for centering) */
```

Auto Margin (Horizontal Centering)

```
.container {
  width: 500px;
  margin: 0 auto;
}
```

This centers the container **horizontally** if a fixed width is set.

Margin Collapse

When two vertical margins meet (e.g., margin-bottom of one element and margin-top of the next), the **larger one wins**, not their sum.

```
h1 {
  margin-bottom: 30px;
}

p {
  margin-top: 20px;
}
```

The space between them will be **30px**, not 50px.

Summary

- **Padding** pushes content **inward**.
- **Margin** pushes the element **outward**.
- Use shorthand to simplify your CSS.
- Be aware of **margin collapsing** in vertical spacing.